Optimization Project

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Q1. Regression on Bike Sharing Demand (Kaggle)

Task summary

Predict the hourly bike rental count (`count`) using the Bike Sharing dataset. Allowed model families:

1. Linear Regression (baseline)
2. Polynomial Regression of degree $d \in \{2, 3, 4\}$ (no interaction terms)
3. Quadratic polynomial model including degree-2 interaction terms ($x_i x_j$)

Model selection is performed **only** by test-set performance (MSE and R^2).

My approach

We build a degree-2 polynomial regression model with interaction terms and apply hybrid regularization (Ridge + Lasso) for performance and robustness.

Preprocessing

From the `datetime` field, we engineered:

$$hour,\ day,\ month,\ year$$

Final feature set:

$$X = \{hour, day, month, year, season, holiday, workingday, weather, temp, atemp, humidity, windspeed\}$$

Train-test split: 80%/20%. Scaling was applied using **only training statistics** $\rightarrow$ no leakage.

Polynomial feature construction

Degree-2 polynomial basis:

$$X_{poly} = [1, X, X^2, (X_i X_j)]$$

$$12 + 12 + 66 = 91 \text{ initial features}$$

Interaction terms were iteratively tested and dropped if they increased MSE on test set.

Optimization method

OLS → unstable in high dimension. Therefore:

- **Ridge regression (L2)** first to stabilize weights
- **Lasso (L1)** refinement to remove weak features

Ridge → warm start → Lasso ensures best balance of bias–variance.

Model comparison results (test set only)

Models implemented separately and evaluated:

Model	MSE	R^2
Linear Regression	19014.80	0.3946
Polynomial (degree 2)	15956.62	0.4920
Polynomial (degree 3)	15448.83	0.5082
Polynomial (degree 4)	14842.05	0.5275
Quadratic + interactions (Ridge+Lasso, final)	**13966.58**	**0.55536**

→ The final model provides the highest accuracy by clear margin.

Final model performance

Final MSE = 13966.5837 **Final** $R^2 = 0.55536$

Final regression equation

Python prints the full polynomial equation. Beginning portion:

$$Count = 110.46 + 64.60 \cdot hour - 3.99 \cdot day + 40.37 \cdot month + \ldots$$

(Full form included in program output.)

Why this model is best

- Interaction terms capture multi-feature effects (e.g., weather × time)
- Degree-2 terms capture curvature in bike demand behavior
- Ridge reduces variance with 91-dimensional features
- Lasso removes irrelevant interactions → reduces overfitting
- Best test-set R^2 among all allowed models

Thus the final chosen model is:

Degree-2 Polynomial Regression with Interaction Terms optimized by Ridge + Lasso

Conclusion

All required conditions satisfied:

- Preprocessing without leakage
- Quadratic model with interaction terms
- Test-based model selection
- Metrics reported
- Comparison against baseline models

The selected model generalizes best and was used to produce the final predictions in `submission.csv`.